

and thinking about using the DHCP server on your ADSL modem or Wi-Fi router, that won't work. You'll have to set up a DHCP box (you can install the DHCP server on this machine itself), and set it up as the gateway to the Internet.

As for a DNS server, you can have any DNS servers set up anywhere you want, but they must be able to forward DNS requests to this Samba box.

I'm assuming that the Samba box has an IP address of 192.168.100.1. Replace this with your own IP address (and of course, it goes without saying: the Samba box must have a static IP). Anyway, let's move on to...

Installation

Start by installing all required packages, as follows:

```
$ sudo apt-get install samba4 samba4-clients krb5-user bind9 attr ntp
```

This command will spew out a huge error message, so run it again to make sure all packages (other than *samba4*) are installed properly. Samba 4 is actually fully installed, but the default configuration creation script is broken. This is good, because Samba 4 configuration is pretty involved, and we want to be able to do it from scratch. However, as far as *dpkg* is concerned, the installation failed, and we need to fix that. So we'll tell *dpkg* that *samba4* is installed and fine, by opening up */var/lib/dpkg/status*, searching for the string 'Package: samba4', and next to it, replacing the word 'half-configured' with 'installed'. Then we'll actually configure Samba 4 by running the following commands:

```
$ sudo rm /etc/samba/smb.conf
$ sudo /usr/share/samba/setup/provision --realm=network.local
--domain=NETWORK --adminpass='aBcD1@3*' --server-role=dc
```

This sets up the AD, so there're a couple of things to note here. The *realm* is the Kerberos realm, and is generally the full *domain* name, including the suffix, of your network. The domain is the WINS domain, equivalent to the Workgroup name, and is typically the first name of the domain in upper-case. The *adminpass* is the password for the 'administrator' user (yes, in Samba, *administrator* starts with lower-case 'a', and that's what you need to use when logging in on Windows). The server role is *dc*, which sets up an AD(S) for you.

Watch the output of the provision command carefully. Near the beginning of the output, it'll try and detect which IP address to use. If it detects the IP address of the wrong interface (one that you don't want to serve on), we'll have to change this. Make a note of the wrong IP address it detected—we'll need that later.

Configuration is far from over, but we can now test if Samba 4 is actually working. Run:

```
$ sudo service samba4 start
$ sudo smbclient -L localhost -U%
```

The last command should spit out the following:

Sharename	Type	Comment
-----	----	-----
netlogon	Disk	
sysvol	Disk	
IPC\$	IPC	IPC Service

Congratulations, we've installed Samba 4 successfully. It's time to hook it up to BIND.

Setting up DNS

We're in luck here, as Samba 4 automatically pushes DNS updates out to BIND. All we need to do is to tell BIND to include in its configuration a certain file, and we'll be done. First, though, if a wrong IP has been selected, we need to change it. Do that by running the commands below:

```
$ samba-tool dns update 192.168.100.1 network.local server A
192.168.24.129 192.168.100.1
$ samba-tool dns update 192.168.100.1 network.local @ A
192.168.24.129 192.168.100.1
```

Here, 192.168.100.1 is the IP address of the Samba box, 'server' is the hostname of the box, and 192.168.24.129 is the wrong IP that was detected—so adjust those accordingly. You'll need the administrator password for both commands, but you don't need to run them as the root user.

Then open up */etc/bind/named.conf*, and add the following line:

```
include "/var/lib/samba/private/named.conf";
```

Open */etc/bind/named.conf.options* and add the line given below inside the braces:

```
tkey-gssapi-keytab "/var/lib/samba/private/dns.keytab";
```

If you really have to use AppArmor, you need to add the following rules to */etc/apparmor.d/usr.sbin.named* inside the braces, anywhere towards the end (if you're using a 64-bit build of Ubuntu, replace *i386* with *x86_64*):

```
/var/lib/samba/private/** rkw,
/var/lib/samba/private/dns/** rkw,
/usr/lib/i386-linux-gnu/samba/bind9/** rm,
/usr/lib/i386-linux-gnu/samba/gensec/** rm,
/usr/lib/i386-linux-gnu/lib/modules/ldb/** rm,
/usr/lib/i386-linux-gnu/samba/ldb/** rm,
```

Now do the following:

```
$ sudo service apparmor restart
$ sudo service bind9 restart
```